

INTRODUCTION TO WANDB



By: Ron Sarma

WHAT IS WANDB?

WandB (Weights and Biases) is a powerful ML Operations platform to track, visualize, manage and recover your experiments seamlessly.



Track Everything

Metrics, hyperparameters, model versions, and system resources



Visualize in Real-Time

Interactive dashboards updated during training



Collaborate

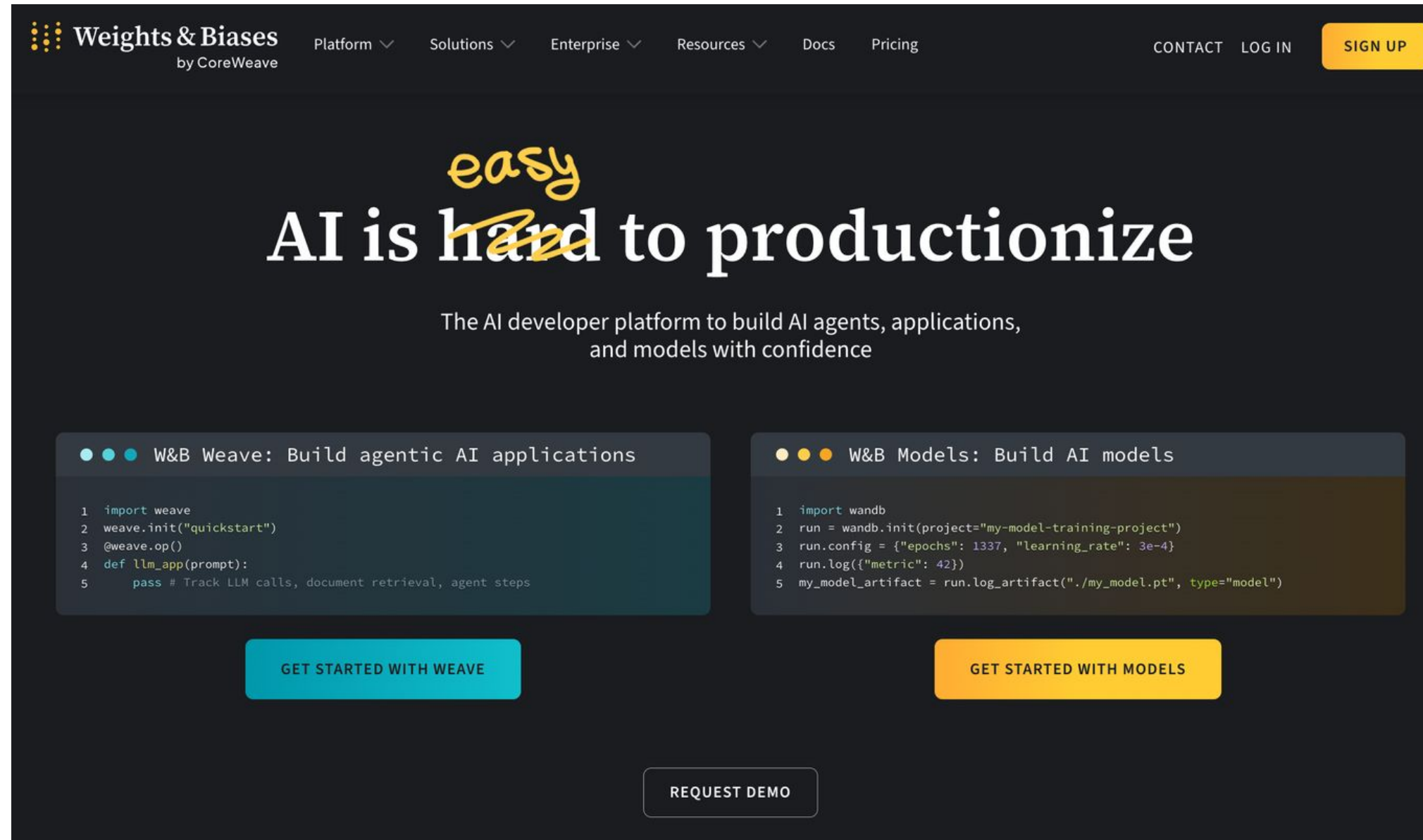
Share experiments with teammates easily



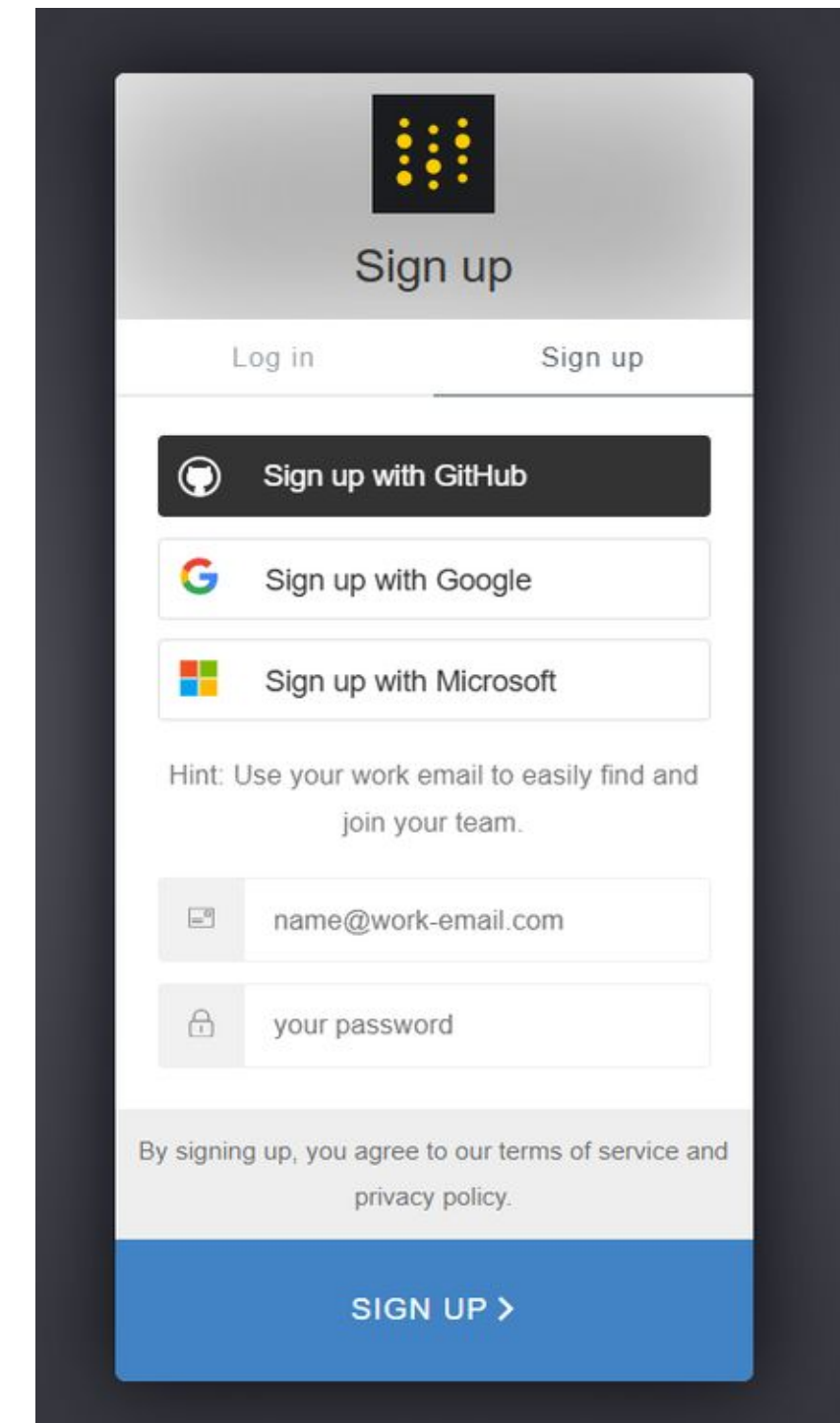
Integrates Seamlessly

Works with PyTorch, TensorFlow, scikit-learn, and more

SET UP WANDB ACCOUNT



<https://wandb.ai/site/>

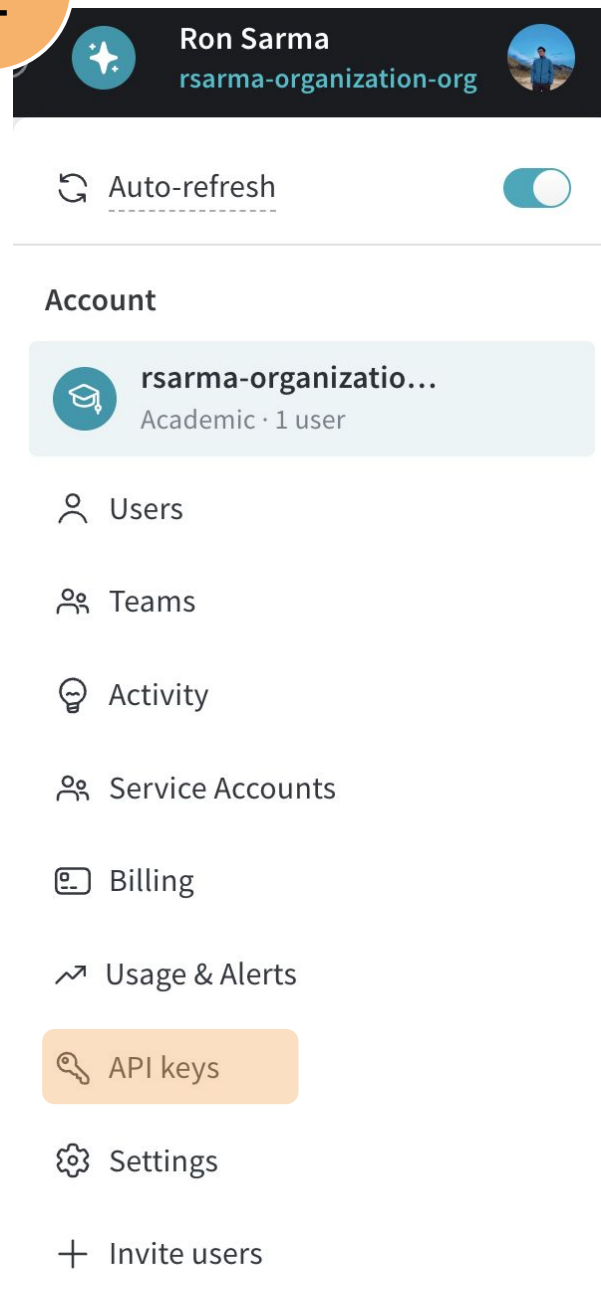


The image shows the "Sign up" form on the W&B website. It features the W&B logo at the top, followed by "Sign up" and "Log in" tabs. Below the tabs are three social login options: "Sign up with GitHub", "Sign up with Google", and "Sign up with Microsoft". A hint says "Hint: Use your work email to easily find and join your team." Below the hint are two input fields: "name@work-email.com" and "your password". At the bottom, there is a checkbox for "By signing up, you agree to our terms of service and privacy policy." and a large blue "SIGN UP >" button.

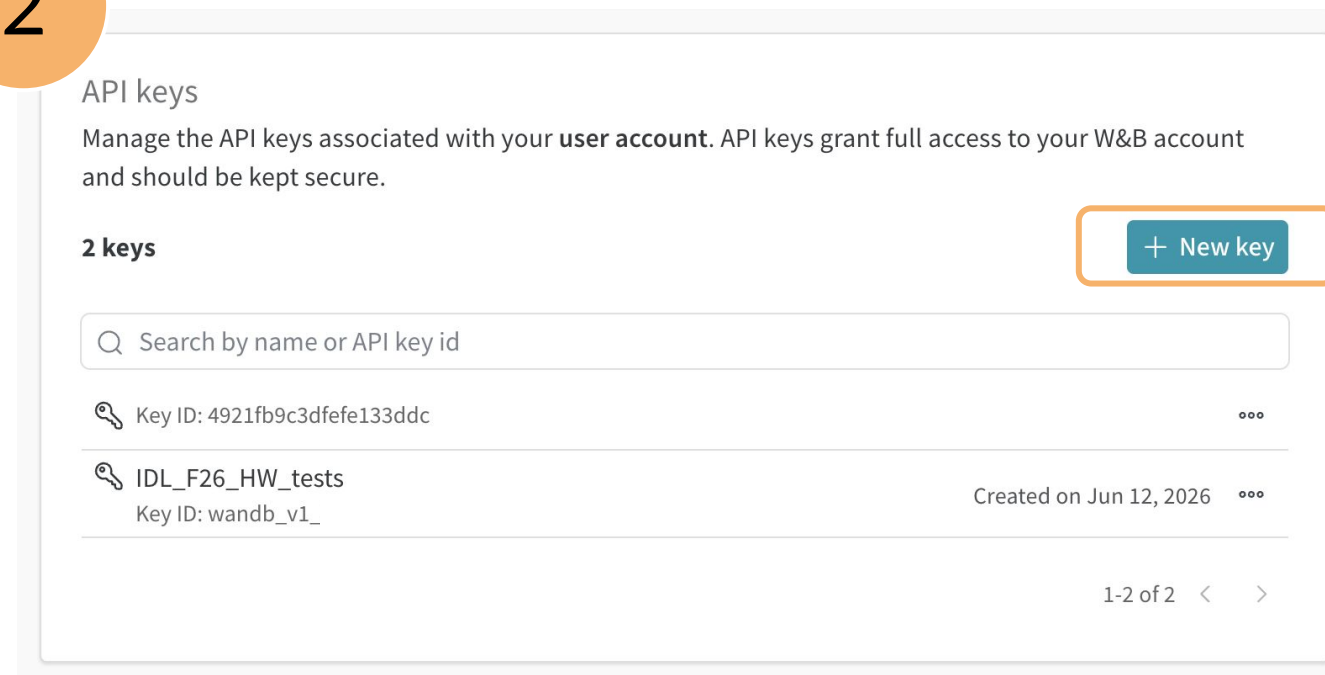
SET UP WANDB API KEY

- Your API key authenticates your machine/notebook so runs can upload to WandB.
- You can find your key in the “API Keys” section in your WandB user settings.

1



2



3

Create new API key

Step 2 of 2 : Save key

⚠️ **SAVE THIS KEY NOW.** This is the only time you'll see the complete key.

API KEY Copy

wandb_v1_
ev 3p

Key information

Key ID	: wandb_v1_
Name	: test
Created	: Aug 01, 2026
Owner	: Ron Sarma

Never post your API key in GitHub, Piazza, screenshots, or shared notebooks!

If it gets leaked, please rotate the key immediately in WandB settings

WANDB - NEW PROJECT

1

rsarma



Ron Sarma

rsarma 

rsarma@andrew.cmu.edu

Edit profile

 Carnegie Mellon Univ...

 United States

 ronsarma.github.io/ 

Teams

 rsarma-organization

Profile Reports Projects Stars

Intro



Introduce yourself to the world

Let others know a little bit about who you are and what you do.

Add an intro

Reports



Share insights and showcase your findings

Reports are your brain dumps, eureka moments, and everything in between. They play nicely with everything else in W&B, so you can share, store or explain your findings.

Showcase a report

2

rsarma



Ron Sarma

rsarma 

rsarma@andrew.cmu.edu

Edit profile

Profile Reports Projects Stars

Your starred projects



No starred projects

Star a project from your projects list and it will always show here.

Your rsarma-organization-org projects

 Search by project name

showing 10  

Name	Last Run or Trace	Project Visibility	Runs	Traces & Spans
wandb-quickstart	1 hour ago	 Team	3	0

 New project

3

     Ron Sarma

rsarma-organization-org 

Create a new project 

This project will be created inside your academic account.

Team

rsarma-organization 

Name

test

Project visibility

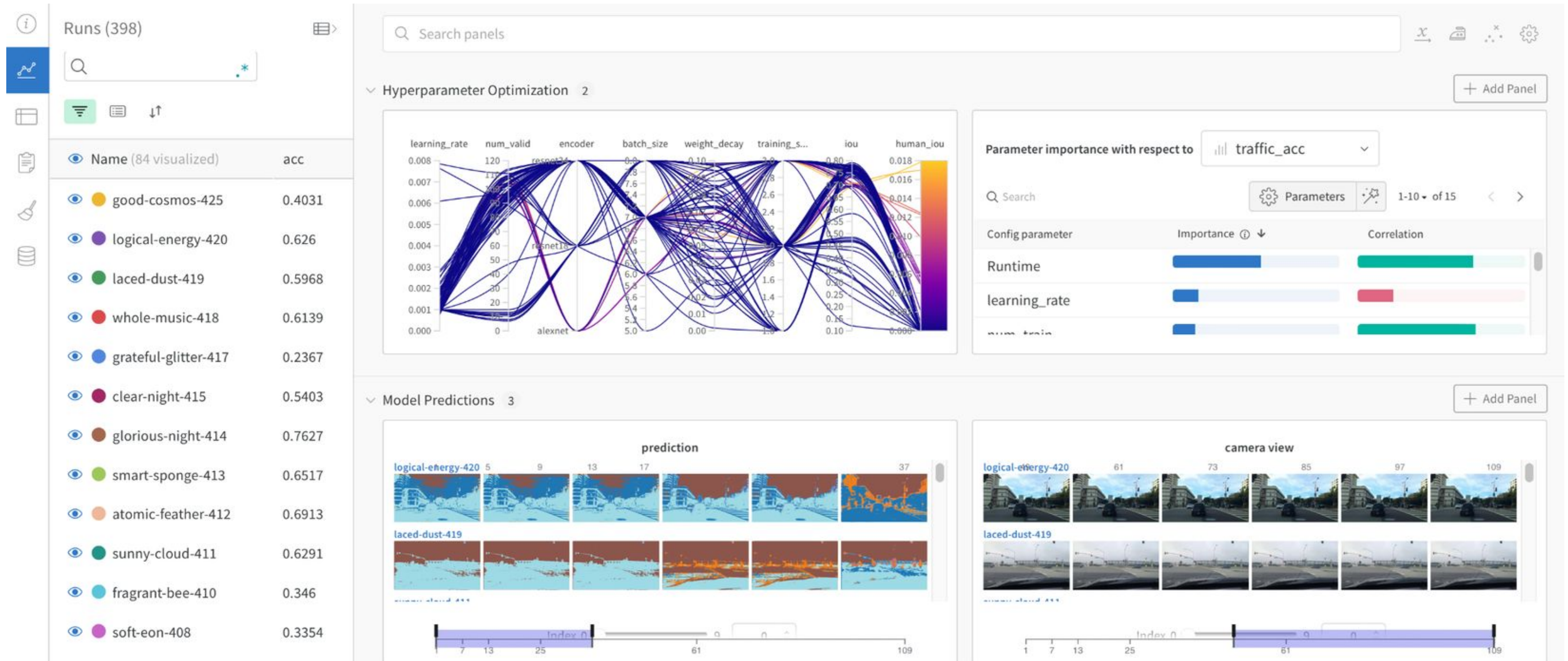
 Team 

Description *(optional)*

potato

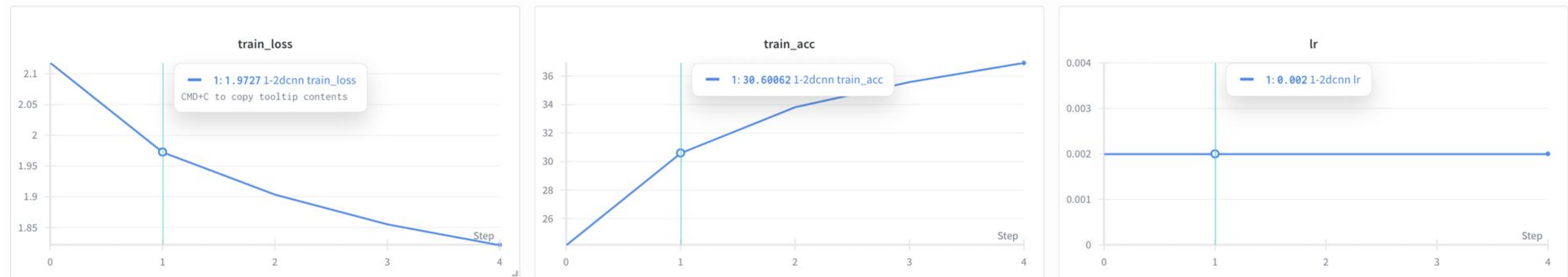
Create project

WANDB DASHBOARD



KEY FEATURES OF WANDB

Real-time Performance Logging



- Visualize metrics like Loss and Accuracy F1 in real-time as the model trains.
- Keep track of different models and training runs.
- Spot overfitting or exploding gradients instantly using interactive charts.

KEY FEATURES OF WANDB

Crash Recovery & Resumption

Rsarma's workspace

Personal workspace

Saved 1 minute ago

Runs 12

Search runs

Filter

Group

Sort

New sweep

Download

Columns

☐ ☒ NAME 12 listed	STATE	NOTES	USER	TAGS	CREATED	RUNTIME	SWEEP	NAME	BATCH_SIZE	DECC
☐ ☒ this_it_it	Finished	Add notes...	rsarma		9mo ago	4d 21h 6m 15s	-	Ron	128	0.2
☐ ☒ well_th...g_again	Finished	Add notes...	rsarma		9mo ago	6h 22m 22s	-	Ron	128	0.2
☐ ☒ this_is_...noying	Crashed	Add notes...	rsarma		9mo ago	1h 29m 40s	-	Ron	128	0.2
☐ ☒ this_is_...noying	Finished	Add notes...	rsarma		9mo ago	14h 16m 40s	-	Ron	128	0.2

- WandB can save your model's checkpoints to the cloud.
- By passing the specific Run ID, you can download the weights and resume the epoch exactly where you left off.

KEY FEATURES OF WANDB

Collaboration

- All team members log runs to the same "Project" bucket.
- See everyone's runs side-by-side to determine whose model is performing best.

Hyperparameter Tuning

- Automating the search for the best config (Learning Rate, Batch Size, Optimizer).
- Let WandB try different configurations while you sleep!

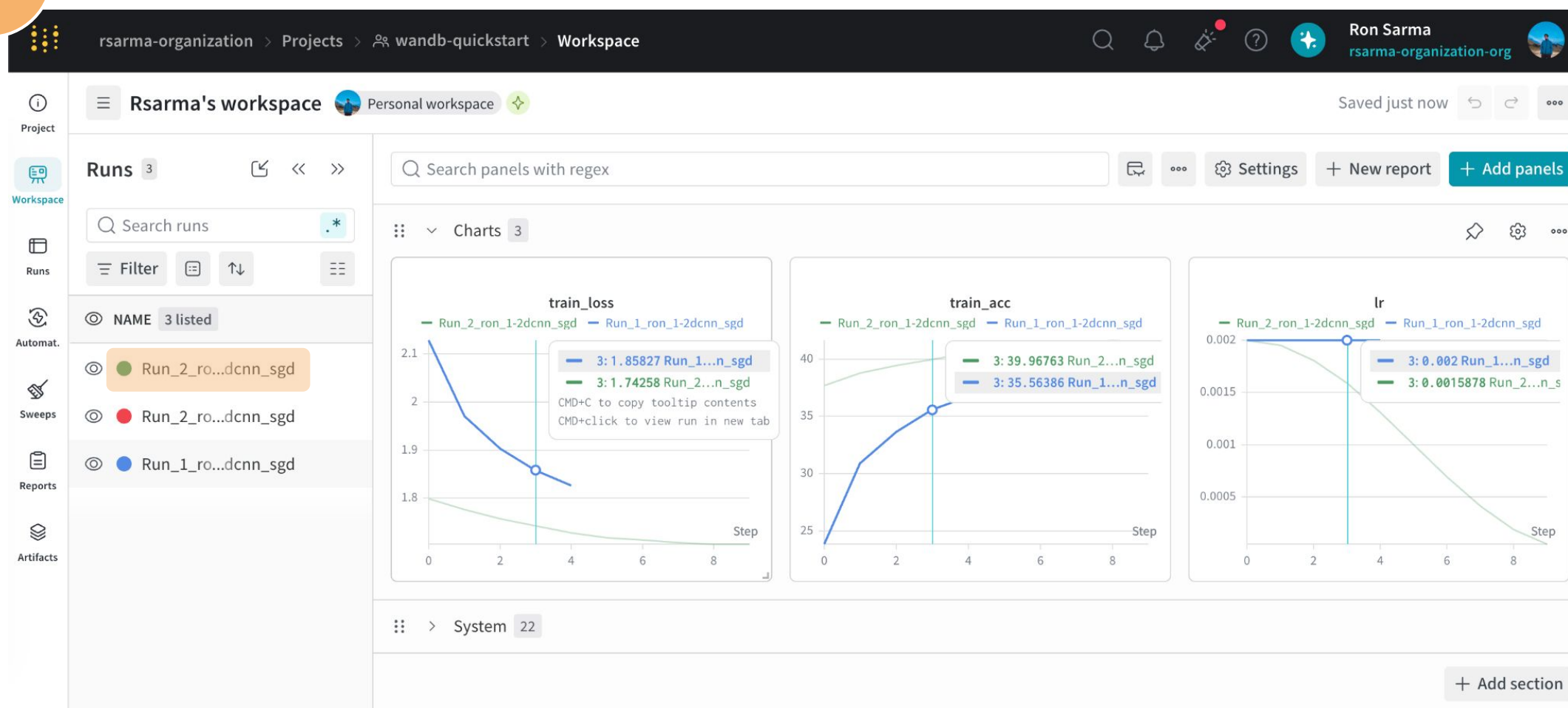
INSTALLATION & NOTEBOOK SET UP

Notebook Demonstration

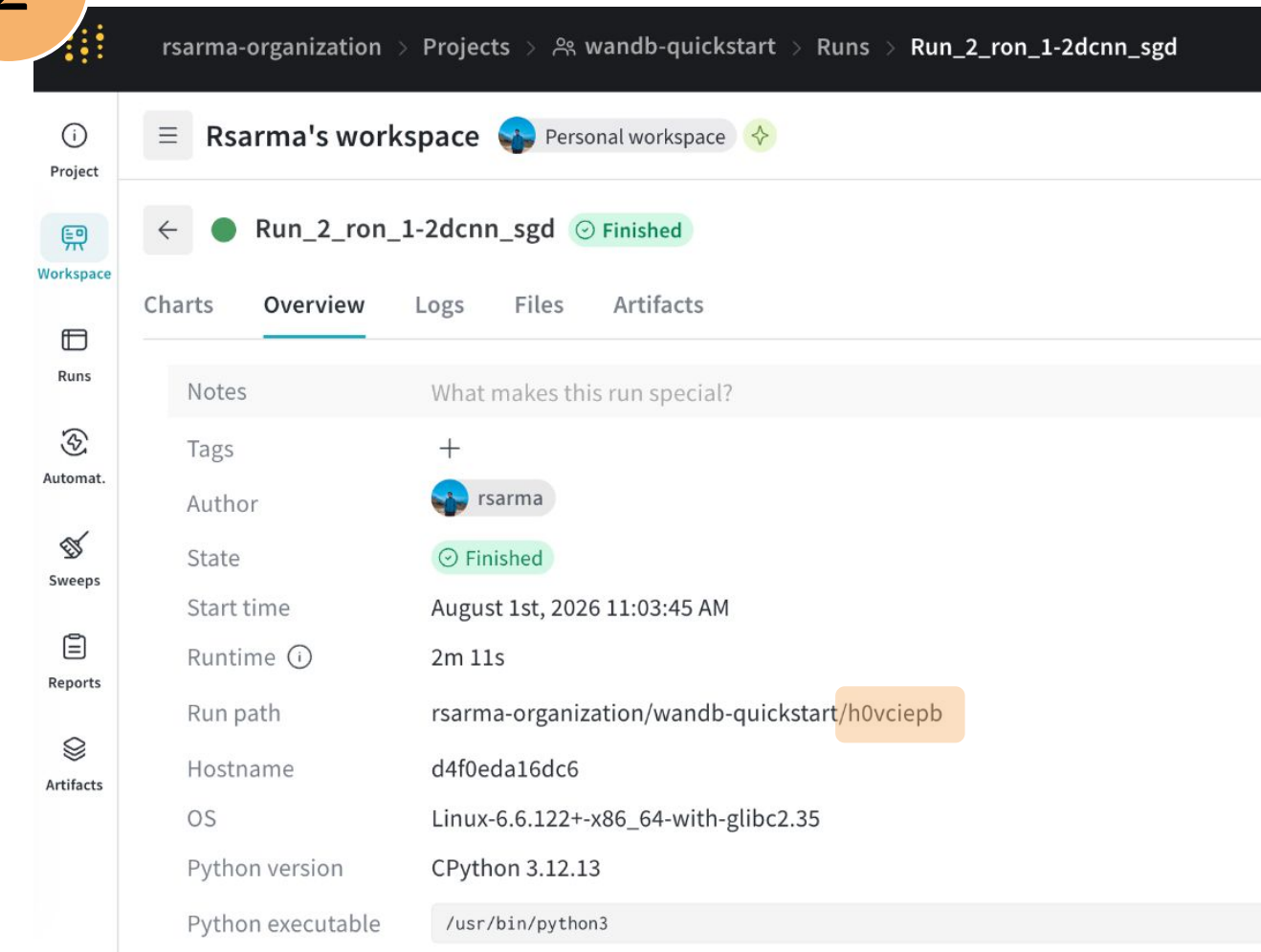
Credits: Fall 2024 TAs + Akshara (S26)

ACCESSING YOUR RUNS

1



2



WANDB STORAGE FOR ACCOUNTS

Once your WandB free storage is full, you won't be able to access your runs!

- Individual Account:
 - Provides **100GB** free
 - Can be accessed through <https://wandb.ai/subscriptions>
- Group Account:
 - Provides **5GB** free per team
 - Can be accessed through: <https://wandb.ai/account-settings/<Team-Name>/billing>

THANK YOU!